

DIRECTORY	
IRRIGATION SCHEDULE	IR1.0
IRRIGATION NOTES	IR1.0
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IRRIGATION DETAILS	IR1.3 - IR1.5

IRRIGATION CONSTRUCTION NOTES

1.

DRAWINGS AND BASE INFORMATION - ALL BASE AND PLANTING INFORMATION HAVE BEEN PROVIDED BY MEURAN DESIGN GROUP AND ALLRED & ASSOCIATES. THE CONTRACTOR IS RESPONSIBLE TO NOTIFY HYDROSYSTEMS*KDI OF ANY DISCREPANCIES BETWEEN THE UTILITY OR PLANTING PLANS AND THE IRRIGATION PLAN. IF CONTRACTOR FAILS TO NOTIFY HYDROSYSTEMS*KDI AND MAKES CHANGES TO THE IRRIGATION SYSTEM DESIGN, THEY ASSUME ALL COSTS AND LIABILITIES ASSOCIATED WITH THOSE FIELD CHANGES. REFER TO SPECIFICATIONS FOR ADDITIONAL PROJECT REQUIREMENTS. CONTACT IRRIGATION CONSULTANT FOR CURRENT SPECIFICATIONS IF NOT PROVIDED.
2.

SYSTEM PRESSURE - HYDROSYSTEMS*KDI HAS CONTACTED THE LOCAL WATER DISTRICT THAT SERVES THIS SITE AND THEY HAVE BEEN TOLD THAT THE STATIC WATER PRESSURE IN THIS AREA SHOULD BE 71 PSI. THE CONTRACTOR IS RESPONSIBLE TO FIELD VERIFY PRESSURE PRIOR TO COMMENCING ANY CONSTRUCTION AND NOTIFY HYDROSYSTEMS*KDI OF ANY VARIANCE FROM THE STATED PRESSURE IMMEDIATELY. WRITTEN DOCUMENTATION OF PRESSURE TEST AND RESULTS SHALL BE PROVIDED TO HYDROSYSTEMS*KDI AT CONSTRUCTION ONSET. IF CONTRACTOR FAILS TO FIELD VERIFY PRESSURE AND/OR NOTIFY HYDROSYSTEMS*KDI OF ANY VARIATIONS FROM THIS PRESSURE, THEN THEY ASSUME ALL CONSTRUCTION AND ENGINEERING COSTS ASSOCIATED WITH SYSTEM MODIFICATIONS REQUIRED TO ACCOMMODATE ACTUAL SITE PRESSURE. THIS SYSTEM HAS BEEN DESIGNED FOR A REQUIRED STATIC PRESSURE OF 71 PSI MINIMUM.
3.

IRRIGATION SYSTEM OPERATION INTENT - THIS IRRIGATION SYSTEM HAS BEEN DESIGNED TO IRRIGATE THE ESTABLISHED LANDSCAPE WITHIN A THREE NIGHT PER WEEK, SIXTEEN HOUR PER NIGHT WATERING WINDOW. ESTABLISHMENT WATERING WILL REQUIRE UP TO TWICE AS MUCH IRRIGATION FOR A FOUR TO SIX WEEK PERIOD. THE DESIGN IS BASED ON THE FOLLOWING PROJECTED WEEKLY APPLICATION RATES AFTER ESTABLISHMENT. THESE FIGURES ARE BASED ON A 30-YEAR AVERAGE WEATHER DATA AND WILL NEED TO BE ADJUSTED DUE TO SEASONAL CHANGES AND WEATHER CONDITIONS ABOVE AND BELOW THE AVERAGE VALUES UTILIZED.

BLUEGRASS TURF

2.23" PER WEEK PEAK SEASON

ORNAMENTAL PLANTINGS

0.89" PER WEEK PEAK SEASON

NATIVE SEED MIXES

0.74" PER WEEK PEAK SEASON (TWO SEASONS)

NOTE: IT IS THE INTENT OF THIS DESIGN THAT NATIVE AREAS WOULD ONLY BE IRRIGATED FOR ESTABLISHMENT. SYSTEM WILL REMAIN FOR USE DURING YEARS WITH LESS THAN NORMAL RAINFALL.
6.

EQUIPMENT INSTALLATION - IT IS THE INTENT OF THIS DESIGN THAT ALL IRRIGATION EQUIPMENT BE INSTALLED WITHIN PROPERTY LIMITS AND WITHIN LANDSCAPED AREAS. INSTALLATION SHALL BE COORDINATED WITH OTHER UTILITY WORK. ALL OTHER UTILITIES SHALL TAKE PRECEDENCE OVER IRRIGATION LOCATION. ANY EQUIPMENT OTHER THAN VALVE BOXES OR SLEEVING THAT CONTAINS PIPE OR WIRES SHOWN OUTSIDE OF THESE LIMITS IS SHOWN IN THAT LOCATION FOR GRAPHICAL CLARITY ONLY. ALL VALVE BOXES SHALL BE INSTALLED A MINIMUM OF 2'-0" FROM EDGE OF ANY PAVED SURFACES UNLESS SPECIFICALLY INDICATED ON PLANS. BOXES INSTALLED IN OPEN TURF AREAS SHALL BE KEPT TO EDGES AND STAKED FOR REVIEW IF ALONG HIGH TRAFFIC AREAS. ALL VALVE BOXES SHALL BE PLACED A MINIMUM OF 3'-0" FROM THE CENTERLINE OF ANY DRAINAGE SWALE. ALL VALVE BOXES WITHIN PAVEMENT SHALL BE TIER 15 RATED BOXES FOR HEAVY DUTY NON-DELIBERATE TRAFFIC. BOX LID COLOR SHALL MATCH ADJACENT MATERIALS, I.E. GREEN IN TURF, TAN IN WOOD MULCH, GRAY IN STONE MULCH, PURPLE FOR RECLAIMED WATER SYSTEMS (IF REQUIRED). REFER TO LANDSCAPE PLANS FOR MATERIAL COLORS AND TYPES. ALL BOXES SHALL BE INSTALLED TO BE FLUSH WITH GRADE AND IN AN ORDERLY MANNER. WHERE MORTAR PAVING LIDS ARE INSTALLED ABOVE BOXES, IRRIGATION BOX WITH LID SHALL BE LOWERED TO ACCOMMODATE PAVING LID. REFER TO LANDSCAPE FOR ADDITIONAL INFORMATION, TO BE INSTALLED PER MANUFACTURE RECOMMENDATIONS.
7.

PIPING INSTALLATION - IRRIGATION PIPING SHALL MAINTAIN A MINIMUM DISTANCE FROM BUILDING FOUNDATIONS OF 5 FEET OR AS DESCRIBED IN SOILS REPORT, WHICHEVER IS GREATER. NO SPRAY IRRIGATION SHALL OCCUR WITHIN 10 FEET OF THE FOUNDATION. NO DRIP IRRIGATION SHALL OCCUR WITHIN 5 FEET OF THE FOUNDATION UNLESS SOIL MOISTURE SENSORS ARE INSTALLED ON VALVES SERVICING THESE AREAS. ALL IRRIGATION PIPING AND EMISSION DEVICES LOCATED ON TOP OF OR WITHIN BUILDING STRUCTURE SHALL CONFORM TO WATERPROOFING CONSULTANT REQUIREMENTS. PIPE ROUTING MAY BE SHOWN WITHIN THESE DISTANCES FOR GRAPHICAL CLARITY ONLY.
8.

MANUAL DRAIN VALVES - CONTRACTOR TO INSTALL ONE MANUAL DRAIN VALVE ON PRESSURE SUPPLY LINE DIRECTLY DOWNSTREAM OF BACKFLOW PREVENTER AND AT ALL LOW POINTS AND DEAD ENDS OF PRESSURE SUPPLY PIPING TO ENSURE COMPLETE DRAINAGE OF SYSTEM. CONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING THESE LOCATIONS IN-FIELD AND INSTALLATION LOCATIONS SHALL BE NOTED ON AS-BUILTS.
9.

POP-UP SPRAY NOZZLES - CONTRACTOR TO INSTALL PLASTIC NOZZLES ON ALL POP-UP SPRAY HEADS. INSTALL 15 SERIES NOZZLES ON ALL HEADS SPACED AT 12' TO 14'. INSTALL 12 SERIES NOZZLES ON ALL HEADS SPACED 10' TO 11'. INSTALL 10 SERIES NOZZLES ON ALL HEADS SPACED AT 8' TO 9'. INSTALL 8 SERIES NOZZLES ON ALL HEADS SPACED AT 7'. INSTALL 5' NOZZLES ON ALL HEADS SPACED AT 5'. INSTALL SIDE STRIP NOZZLES ON ALL HEADS WITH AN 'S' DESIGNATION AND RIGHT AND LEFT CORNER STRIP NOZZLES ON ALL HEADS WITH AN 'L' OR 'R' DESIGNATION. VARIABLE ARC NOZZLES SHOULD BE UTILIZED ADJACENT TO CURVILINEAR SHRUB BEDS OR FOR ANY ANGLES THAT ARE NOT A STANDARD NOZZLE ANGLE.
10.

UNLABELED PIPING - ALL UNLABELED LATERAL PIPING SHALL BE 1" MINIMUM UNLESS OTHERWISE NOTED.
11.

SLEEVING - ALL SLEEVING UNDER PAVED SURFACES SHOWN ON PLANS IS BY CONTRACTOR UNLESS OTHERWISE NOTED. SLEEVING SHALL BE INSTALLED IN THE SIZES AND QUANTITIES SHOWN ON PLANS OR BASED ON THE SCHEDULE BELOW. WHERE SLEEVES ARE SHOWN, BUT NOT LABELED, FOLLOW THE SCHEDULE BELOW. ALL MAINLINE, CONTROL WIRES AND DRIP LINES UNDER PAVED SURFACES ARE TO BE INSTALLED IN SLEEVING. ALL MAINLINE SLEEVE LOCATIONS TO INCLUDE A SEPARATE WIRE SLEEVE.

SLEEVED PIPE SIZE/WIRE QUANTITY

REQUIRED SLEEVE SIZE & QUANTITY

3/4" - 1 1/4" PIPING

2" PVC (1)

1 1/2" - 2" PIPING

4" PVC (1)

2-WIRE CABLE

2" PVC (1)

12.

2-WIRE SYSTEM NOTES - CONTRACTOR SHALL INSTALL ALL TWO-WIRE COMPONENTS PER MANUFACTURER'S RECOMMENDATIONS AND STANDARDS.

12.1.

CONTRACTOR SHALL USE ONLY MANUFACTURED 2-WIRE DECODER CABLE (SEE SCHEDULE FOR SPECIFIC 2-WIRE CABLE).

12.2.

ONLY USE SINGLE STATION DECODERS (SEE SCHEDULE FOR SPECIFIC MODEL).

12.3.

LOOP 5' OF 2-WIRE DECODER CABLE INTO ALL VALVE BOXES (WITH DECODERS AND SPLICES) FOR MAINTENANCE.

12.4.

LOOP 2' OF 2-WIRE DECODER CABLE AS AN EXPANSION LOOP AT ALL CHANGES OF DIRECTION.

12.5.

USE ONLY 3M DBR-6 WATERPROOF CONNECTORS ON ALL WIRE SPLICES AND ALL WIRE SPLICES ARE TO BE MADE WITHIN A VALVE BOX WITH CONTROL VALVES OR A SEPARATE 10" ROUND VALVE BOX FOR WIRE SPLICES.

12.6.

INSTALL SURGE PROTECTOR RODS OR PLATES 8 LF. PERPENDICULARLY FROM VALVES, DECODERS, AND COMMUNICATION WIRE. WHERE 8 LF IS NOT AVAILABLE, REFER TO ASIC GUIDELINES FOR PLATE LAYOUT.

12.7.

GROUNDING SHALL BE IN AN IRRIGATED AREA.

12.8.

GROUND ALL DECODERS AND DECODER WIRE A MINIMUM OF EVERY 600' OF WIRE AND AT ALL ENDS OF 2-WIRE DECODER CABLE RUN.

13.

ADJUSTMENT - CONTRACTOR SHALL FINE TUNE/ADJUST THE IRRIGATION SYSTEM TO REDUCE/AVOID OVERSPRAY ONTO HARD SURFACES BY ADJUSTING NOZZLE DIRECTION AND NOZZLE RADIUS.

14.

PLANS AND SPECIFICATIONS - CONTRACTOR RESPONSIBLE TO ENSURE WORK CONFORMS TO PLANS AND SPECIFICATIONS. AT ONSET OF CONSTRUCTION, VERIFY PLANS ARE CURRENT. DO NOT CONSTRUCT ANY PORTIONS OF THE IRRIGATION OFF OF PLANS MARKED NOT FOR CONSTRUCTION OR FOR REVIEW ONLY. FAILURE TO VERIFY CURRENT PLANS MAY RESULT IN RECONSTRUCTION AT CONTRACTOR'S EXPENSE. WHERE REQUIRED BY CITY, TOWN OR WATER DISTRICT ENTITY, CONTRACTOR SHALL CONSTRUCT ONLY OFF PLANS STAMPED WITH APPROVAL. REVISIONS TO APPROVED OR STAMPED PLANS SHALL CONFORM TO FIELD CHANGE PROCEDURES AND DOCUMENTATION.

15.

16.

SIMULTANEOUS ZONE OPERATION - THIS IRRIGATION SYSTEM HAS BEEN DESIGNED TO OPERATE MULTIPLE ZONES SIMULTANEOUSLY BASED ON INDIVIDUAL ZONE FLOW. THE DESIGN IS INTENDED TO OPERATE MULTIPLE VALVES, UP TO THE MAXIMUM FLOW IN THE POINT OF CONNECTION NOTE. REFER TO CONTROLLER SPECIFICATION FOR MAXIMUM SIMULTANEOUS VALVE COUNT.

17.

BACKFLOW DEVICES: ALL CONNECTION COMPONENTS AND BACKFLOW DEVICES SHALL BE LEAD-FREE. CONTRACTOR SHALL CONTACT WATER SERVICE PROVIDER FOR ANY ADDITIONAL REQUIREMENTS REGARDING BACKFLOW TESTING.

18.

CITY DETAILS - HYDROSYSTEMS-KDI WILL BE RESPONSIBLE FOR THE IRRIGATION DESIGN ITSELF. DETAILS #2 & #6 ARE STANDARD DETAILS DESIGNED AND SPECIFIED BY THE APPLICABLE GOVERNING AUTHORITY, CITY OF WESTMINSTER. HYDROSYSTEMS*KDI DID NOT DESIGN AND SPECIFY THESE DETAILS. HYDROSYSTEMS*KDI IS NOT RESPONSIBLE, AND ACCEPTS NO RESPONSIBILITY, FOR THE SELECTION AND SPECIFICATION OF THESE DETAILS, AND HYDROSYSTEMS*KDI MAKES NO REPRESENTATIONS WITH REGARD TO THESE DETAILS.

19.

WATER BUDGETS AND PROJECTIONS - HYDROSYSTEMS-KDI HAS BASED THE IRRIGATION DESIGN AND THE ASSOCIATED PROJECTED WATER USE UPON SUCH FACTORS AS CITY OR WATER DISTRICT IMPOSED REQUIREMENTS, PUBLISHED PLANT SPECIES WATER NEEDS, SELECTED IRRIGATION METHOD EFFICIENCIES AS REPORTED BY INDEPENDENT TESTING FACILITIES, HISTORICAL WEATHER DATA FOR THE PROJECT LOCATION, AND PROPER MAINTENANCE PROCEDURES. HYDROSYSTEMS*KDI IS NOT RESPONSIBLE, AND ACCEPTS NO RESPONSIBILITY, FOR THE ACTUAL WATER USAGE VARIATION THAT IS A RESULT OF FIELD MODIFICATIONS TO THE SYSTEM NOT MATCHING CONSTRUCTION DOCUMENTS, IMPROPER MAINTENANCE, WASTE DUE TO SYSTEM DAMAGE OR VANDALISM, OR WEATHER CONDITIONS THAT DEVIATE FROM PUBLISHED 30 YEAR HISTORICAL AVERAGES.

20.

PRESSURE TESTING - CONDUCT MAINLINE TEST IN PRESENCE OF CONSULTANT. ARRANGE FOR PRESENCE OF CONSULTANT 48 HOURS IN ADVANCE OF TESTING. SUPPLY FORCE PUMP AND ALL OTHER TEST EQUIPMENT. COMPRESSED AIR SHALL NOT BE USED FOR PRESSURE TESTING SYSTEM.

20.1

AFTER BACKFILLING, AND INSTALLATION OF ALL CONTROL VALVES, FILL PRESSURE SUPPLY LINE WITH WATER, AND PRESSURIZE TO 40 PSI OVER THE DESIGNATED STATIC PRESSURE OR 120 PSI, WHICHEVER IS GREATER, FOR A PERIOD OF 2 HOURS.

20.2

LEAKAGE, PRESSURE LOSS - TEST IS ACCEPTABLE IF NO LOSS OF PRESSURE IS EVIDENT DURING THE TEST PERIOD.

20.3

LEAKS - DETECT AND REPAIR LEAKS.

20.4

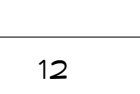
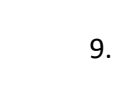
RETEST SYSTEM UNTIL TEST PRESSURE CAN BE MAINTAINED FOR DURATION OF TEST.

20.5

BEFORE FINAL ACCEPTANCE, PRESSURE SUPPLY LINE SHALL REMAIN UNDER PRESSURE FOR A PERIOD OF 48 HOURS.

20.6

PRESSURE TEST SHALL BE SCHEDULED AND PASSED PRIOR TO SCHEDULING OF SUBSTANTIAL COMPLETION WALK-THROUGH.

IRRIGATION SCHEDULE					
SYMBOL	MANUFACTURER	MODEL NO.	DESCRIPTION	DETAIL NO.	
	RAIN BIRD	RD-06-S-P30 WITH U SERIES NOZZLE	POPUP SPRAY HEAD	9	
	RAIN BIRD	RD-12-S-P30 WITH U SERIES NOZZLE	HI-POP SPRAY HEAD	10	
	RAIN BIRD	5006-PL-PC/FC-SAM-R WITH #NOZZLE	GEAR DRIVEN ROTOR	11	
	RAIN BIRD	5012-PL-PC/FC-SAM-R WITH #NOZZLE	HI-POP GEAR DRIVEN ROTOR	12	
	RAIN BIRD	5012-PL-PC/FC-SAM-R WITH #NOZZLE	HI-POP GEAR DRIVEN ROTOR	12	
	RAIN BIRD	5012-PL-PC/FC-SAM-R WITH #NOZZLE	HI-POP GEAR DRIVEN ROTOR	12	
	RAIN BIRD	1402	BUBBLER	13	
	RAIN BIRD / BASELINE	PEB WITH BL-5201	ELECTRIC CONTROL VALVE W/ DECODER	14 & 15	
	RAIN BIRD	44-LRC	QUICK COUPLING VALVE	18	
	BASELINE	BL-3200P W/ BL-GM46-P-VZ CELL MODEM SITE VERIFICATION SERVICE & DBC CERTIFIED START UP SERVICE, WARRANTY TESTING, CALL SPENCER JORDAN DBC 970-416-3521	ELECTRIC CONTROLLER	4 & 5	
	HUNTER	WRF-GLIK	WEATHER SENSOR DEVICE	6	
	FEBCO	825YA	RP BACKFLOW PREVENTER	2	
	STRONG BOX	SBBG-22AL	BACKFLOW PREVENTER ENCLOSURE	3	
		TYPE K COPPER	HARD COPPER PIPE	2	
	OLDCASTLE / CARSON	REFER TO SPECIFICATIONS AND DETAILS	VALVE BOXES	VARIOUS	
	AY MACDONALD	76001 - 1/4 TURN - 1"	MANUAL DRAIN VALVE	17	
		LINE SIZE - 2 1/2" AND SMALLER	GATE VALVE	16	
	RAIN BIRD / BASELINE	PEB WITH BL-5201 (1')	MASTER CONTROL VALVE W/ DECODER	8 & 15	
	BASELINE	BL-PFS100 (1')	FLOW SENSOR W/ DECODER	7 & 15	
		CLASS 200 BE - 2 1/2" & SMALLER	PVC MAINLINE	19	
		CLASS 200 BE	PVC LATERAL	19	
		CLASS 160	PVC SLEEVING	20	
	N/S	BASELINE	BL-5201	VALVE DECODER	15
	N/S	PAIGE	P1012D (FOR RAINBIRD, BASELINE)	2-WIRE DECODER CABLE	VARIOUS
	BASELINE	BL-LA01	SURGE PROTECTION	21	
	BASELINE	BISENSOR	SOIL MOISTURE SENSOR	22 - 24	
	NETAFIM	WM100- 0.1-RS-P	SUBMETER	1	
			CONTROLLER GROUNDING	5	
 CONTROLLER & STATION NO. CONTROL VALVE SIZE					

CITY OF THORNTON
FIRE STATION #8
15343 MONACO STREET
THORNTON, COLORADO 80602

REVISION NAME		DATE
ISSUE DATE:		2024 0927

IRRIGATION NOTES
& SCHEDULE

IR1.0

ALLRED & ASSOCIATES

ARCHITECTURE | PLANNING | INTERIOR DESIGN

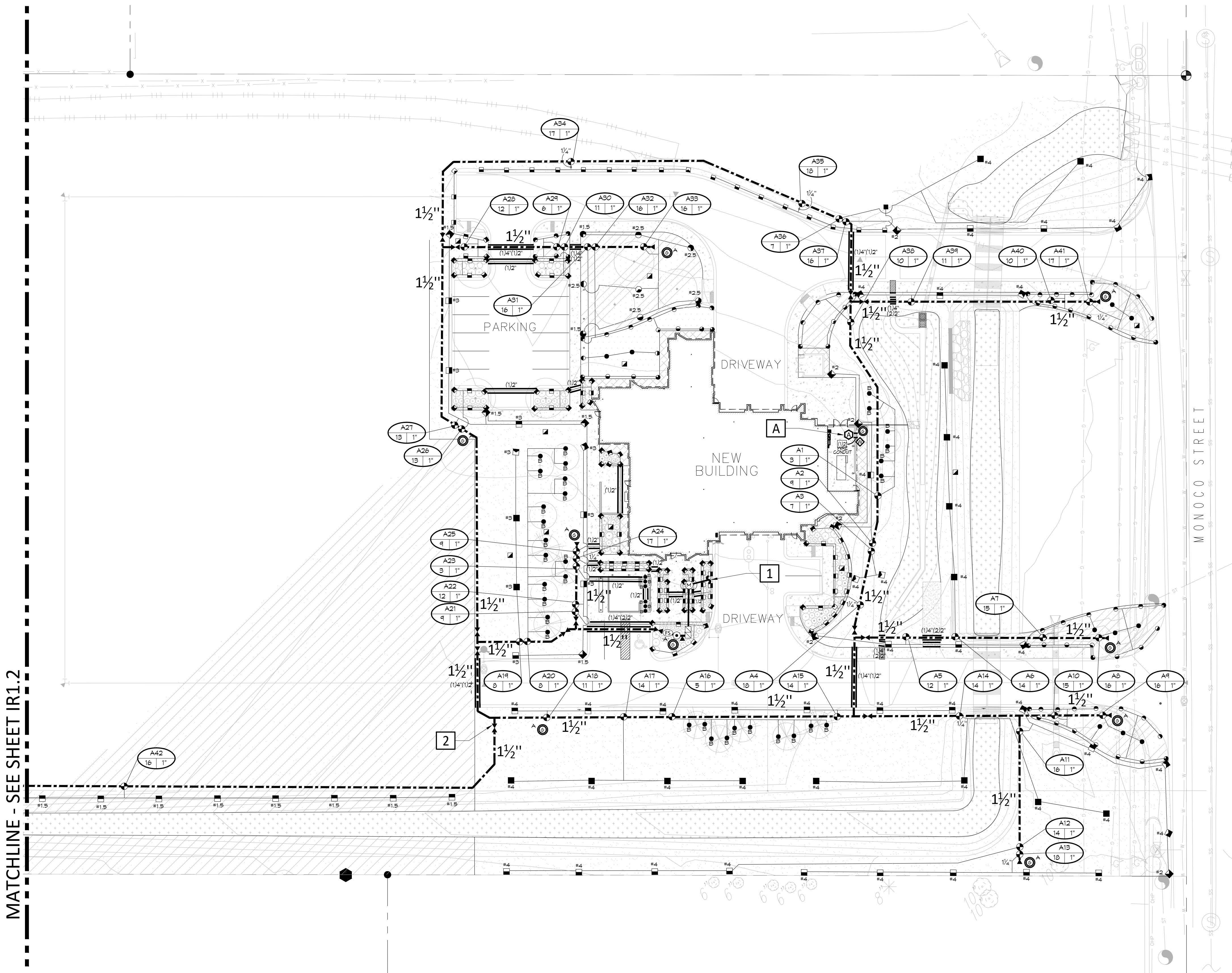
580 BURBANK STREET, SUITE 125, BROOMFIELD, COLORADO 80020

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• JOB NO. 2309

BID SET



MATCHLINE - SEE SHEET IR1.2

- 1

POINT OF CONNECTION #1 - 1"

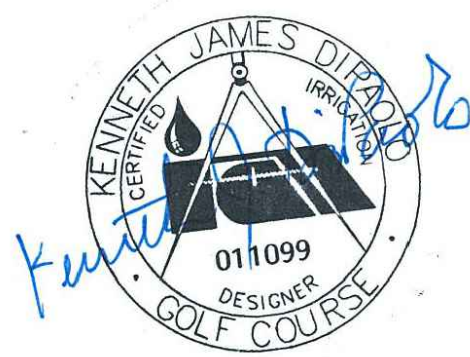
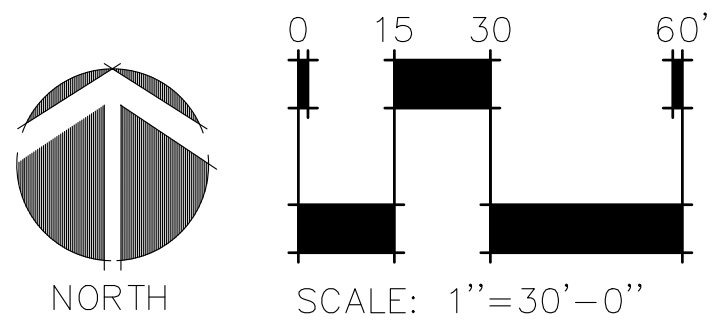
PEAK FLOW REQUIREMENT: 18 GPM. REQUIRED STATIC PRESSURE: 71 PSI

CONTRACTOR TO LOCATE 2" DOMESTIC WATER LINE AT THIS APPROXIMATE LOCATION, DOWNSTREAM OF 2" DOMESTIC WATER METER. EXTEND 1" TYPE K SOFT COPPER AT 54" MINIMUM DEPTH TO BACKFLOW PREVENTER LOCATION. INSTALL ONE 1" SUBMETER, ONE 3/4" WINTERIZATION TEE, ONE 1" REDUCED PRESSURE BACKFLOW PREVENTER WITH PROTECTIVE ENCLOSURE, ONE MANUAL DRAIN VALVE, ONE QUICK COUPLER VALVE, ONE 1" MASTER VALVE, ONE 1" FLOW SENSOR, ONE GATE VALVE, AND EXTEND PVC MAINLINE AS SHOWN. CONNECT FLOW SENSOR AND MASTER VALVE TO ASSOCIATED CONTROLLER VIA TWO-WIRE COMMUNICATION CABLE PER MANUFACTURER'S REQUIREMENTS. SEE DETAIL SHEET FOR REQUIRED PIPE LENGTHS UPSTREAM AND DOWNSTREAM OF FLOW SENSING UNIT. WORK SHALL CONFORM TO LOCAL CODE. FEES AND PERMITS ASSOCIATED WITH WORK ARE TO BE OBTAINED AND PAID FOR BY CONTRACTOR. FINAL BACKFLOW PREVENTER LOCATION SHALL BE APPROVED BY OWNER'S REPRESENTATIVE PRIOR TO INSTALLATION.
- A

CONTROLLER LOCATION "A"

WALL MOUNT ONE CONTROLLER (REFER TO SCHEDULE FOR MODEL & STATION COUNT), REMOTE READY, AT INDICATED LOCATION. 120 VOLT POWER IS AVAILABLE WITHIN TBD LF. OF CONTROLLER LOCATION FROM BUILDING POWER (BY OTHERS - REFER TO ELECTRICAL PLANS). ELECTRICAL METER, WIRE/CONDUIT, STEP-DOWN TRANSFORMER (IF REQUIRED) AND POWER CONNECTION TO CONTROLLER IS BY CONTRACTOR WITH WORK CONFORMING TO LOCAL CODES. FEES AND PERMITS ASSOCIATED WITH WORK ARE TO BE OBTAINED AND PAID BY CONTRACTOR. CONNECT FLOW SENSOR AND MASTER VALVE TO ASSOCIATED CONTROLLER VIA TWO-WIRE COMMUNICATION CABLE PER MANUFACTURER'S REQUIREMENTS. FINAL CONTROLLER LOCATION SHALL BE APPROVED BY OWNER OR OWNER'S REPRESENTATIVE PRIOR TO INSTALLATION. MOUNT ONE WEATHER SENSOR ON ENCLOSURE WALL WHERE THERE IS 20 FT. OF CLEARANCE FROM ANY OVERHANG OR OBSTRUCTING FEATURE. INSTALL WEATHER SENSOR RECEIVER IN CONTROLLER ENCLOSURE. FINAL WEATHER SENSOR LOCATION SHALL BE APPROVED BY OWNER OR OWNER'S REPRESENTATIVE PRIOR TO INSTALLATION.
- 2

CONTRACTOR TO INSTALL ONE 1 1/2" GATE VALVE AND EXTEND NEW 1 1/2" PVC MAINLINE, 2-WIRE CABLE & IRRIGATION, AS SHOWN. THIS MAINLINE & IRRIGATION ON THIS MAINLINE WILL BE TEMPORARY UNTIL FUTURE MAINTENANCE BUILDING IS INSTALLED AT WHICH POINT CONTRACTOR TO REMOVE ALL IRRIGATION ON THIS MAINLINE, GATE VALVE AND CUT & CAP MAINLINE AND 2-WIRE CABLE AT THIS APPROXIMATE LOCATION.



CALL 3 BUSINESS DAYS IN ADVANCE BEFORE YOU DIG. GRADE, OR EXCAVATE FOR THE MARKING OF UNDERGROUND MEMBER UTILITIES.



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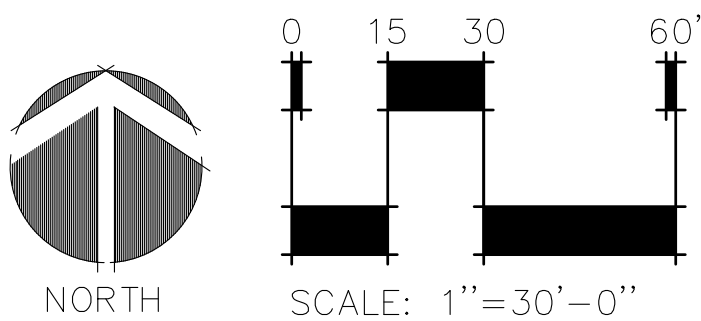
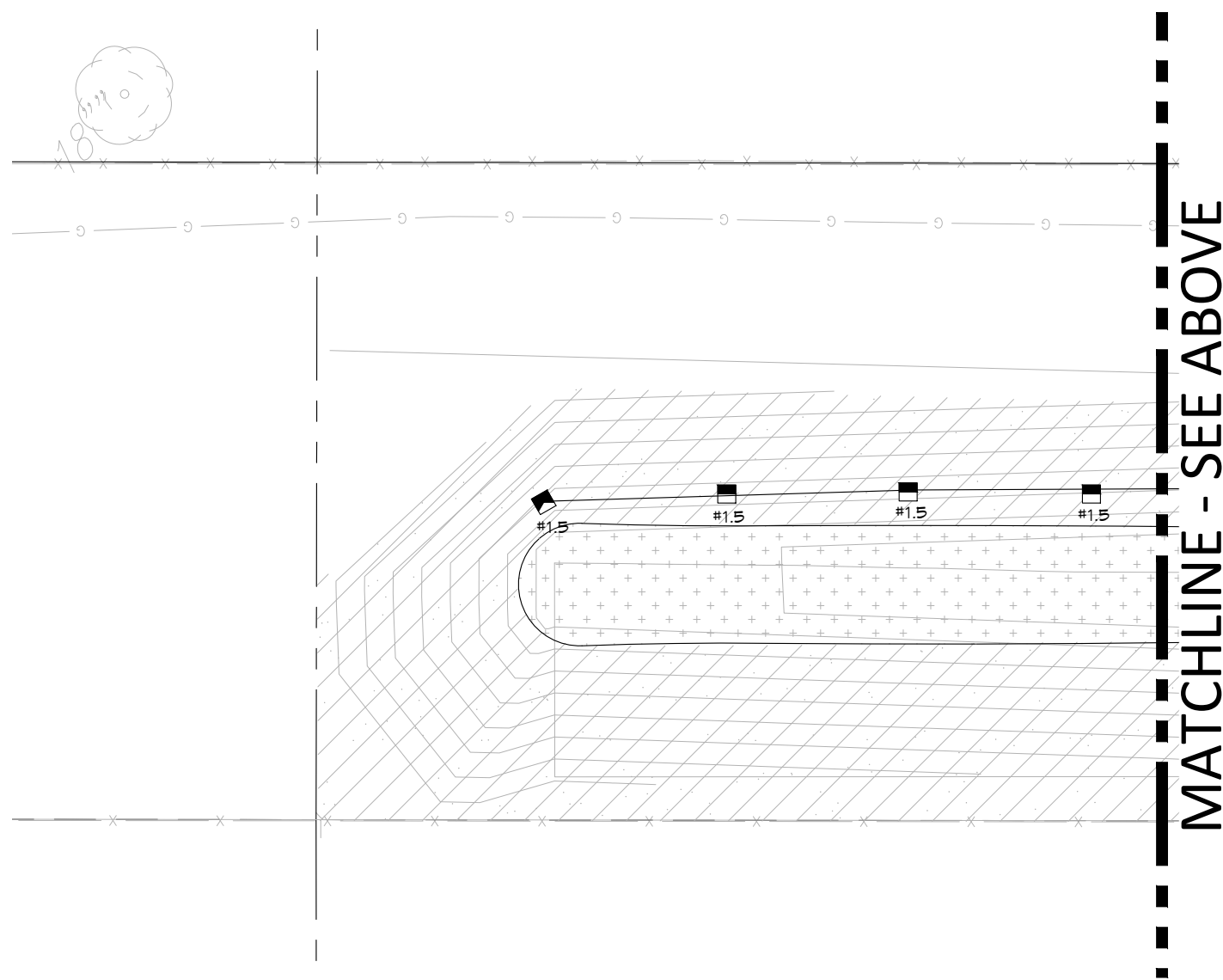
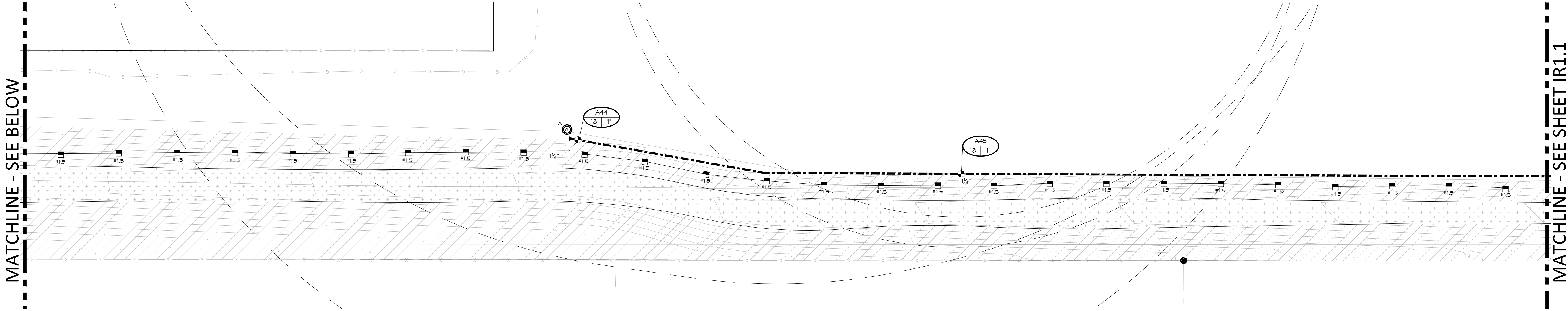
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IRRIGATION PLAN

IR1.1

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IRRIGATION SCHEDULE	IR1.0
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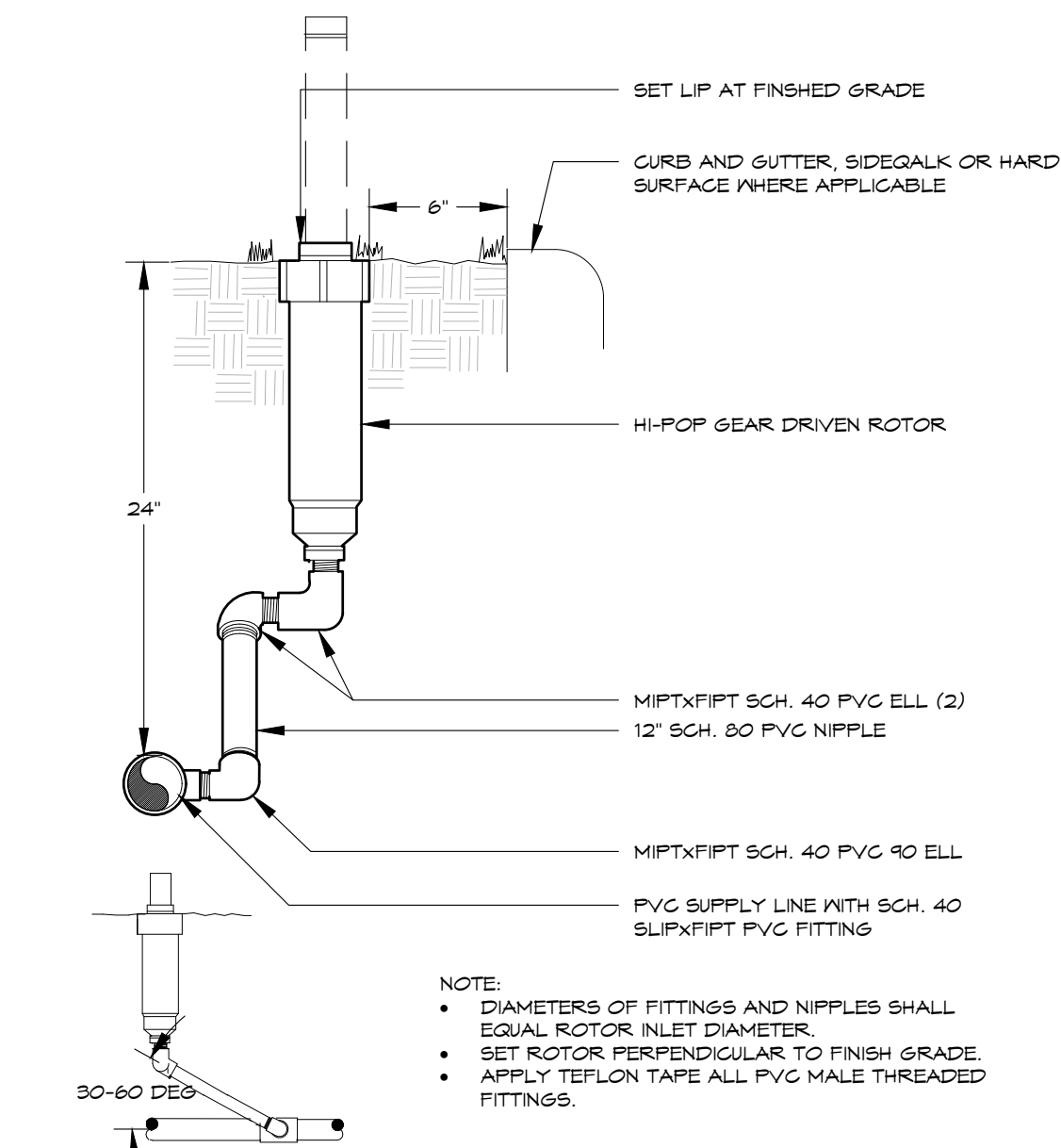


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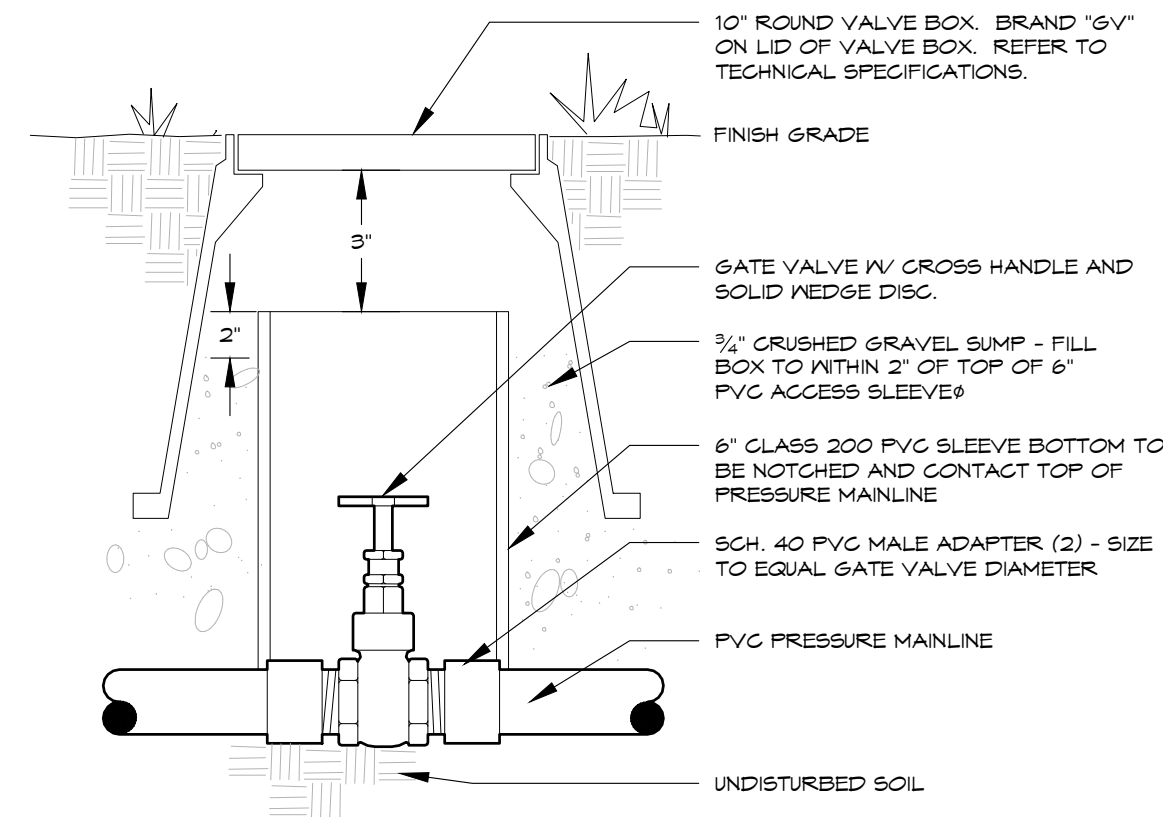
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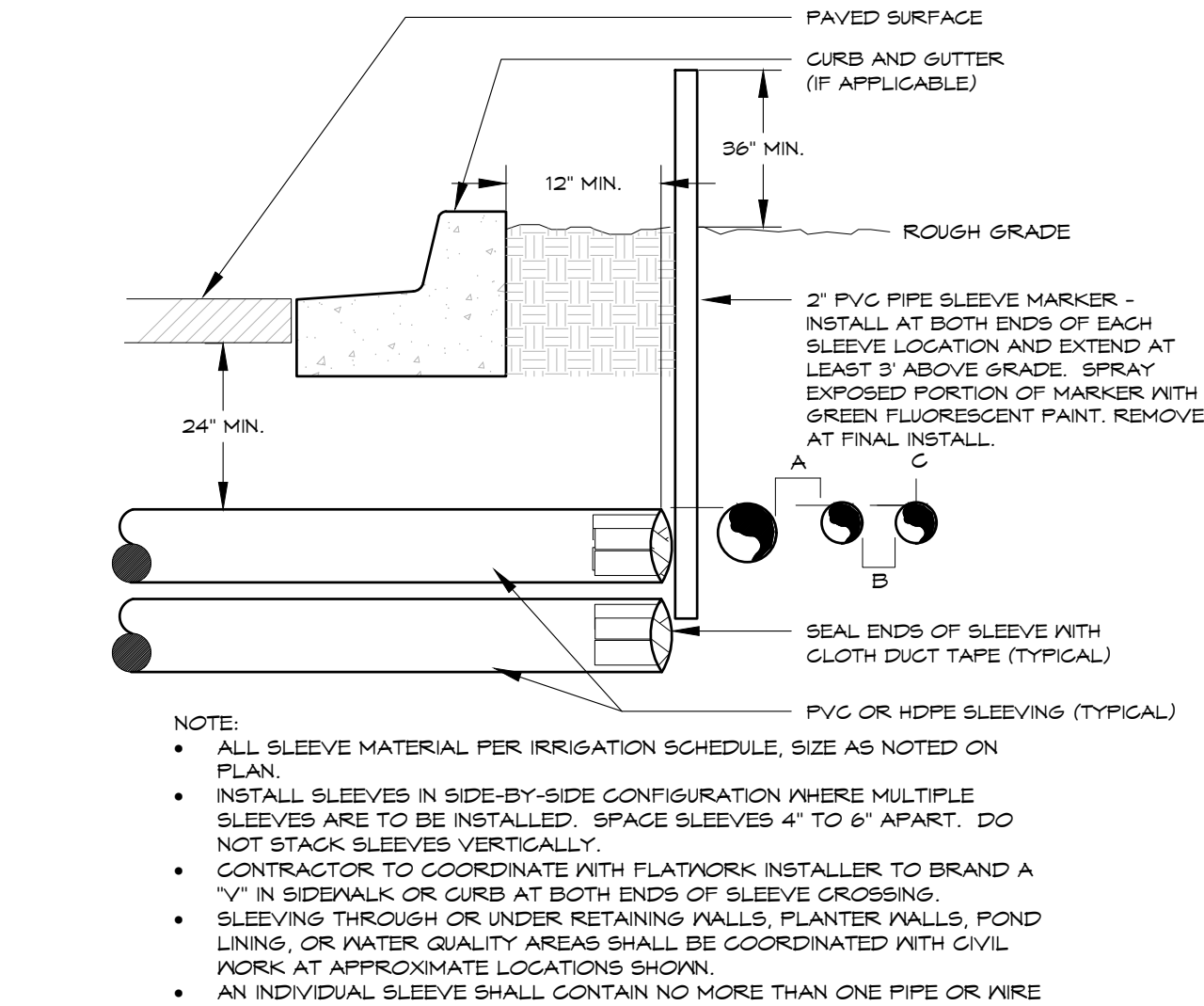
HI-POP GEAR DRIVEN ROTOR
PVC

12



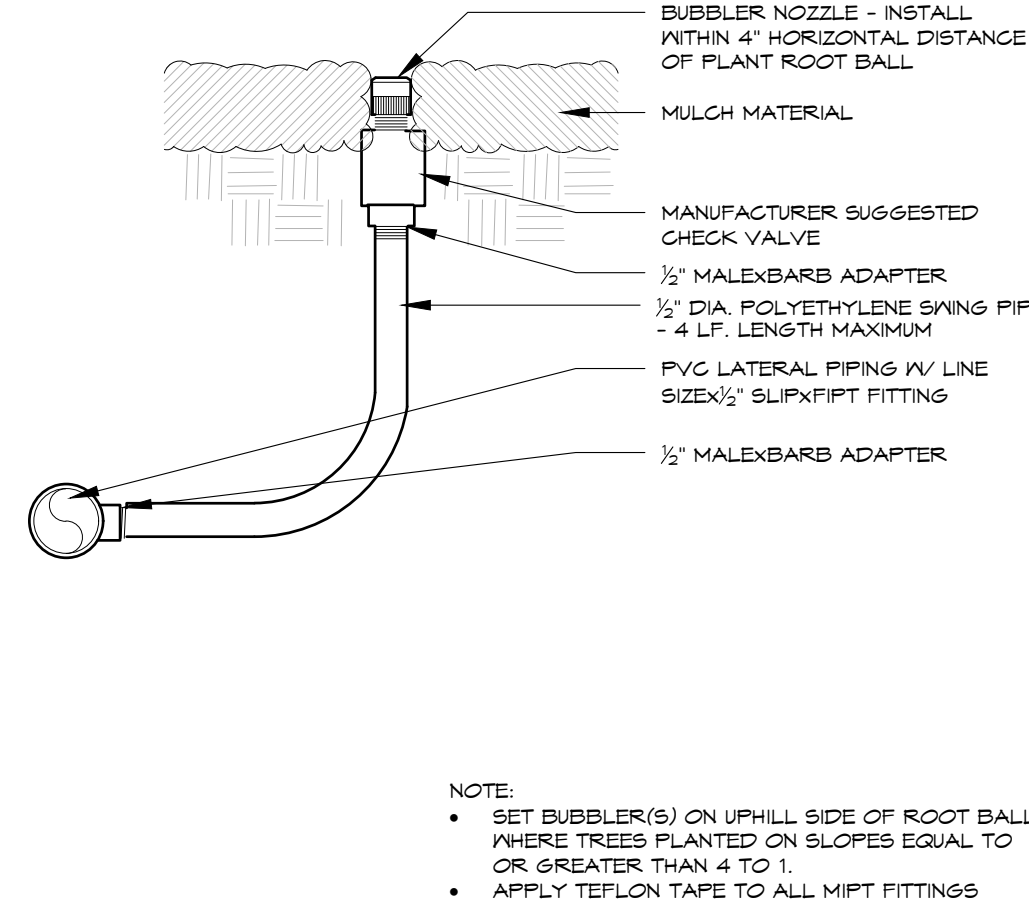
GATE VALVE
2.5" & SMALLER - X-HANDLE

16



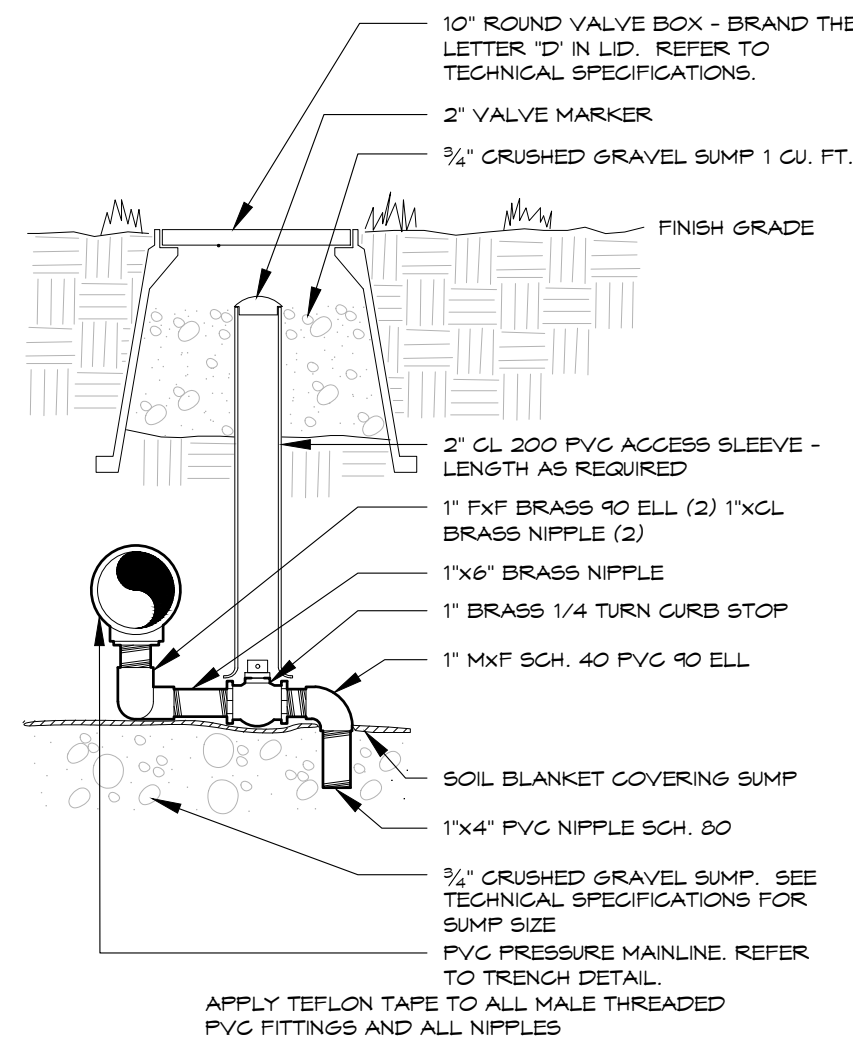
IRRIGATION SLEEVING
TYPICAL

20



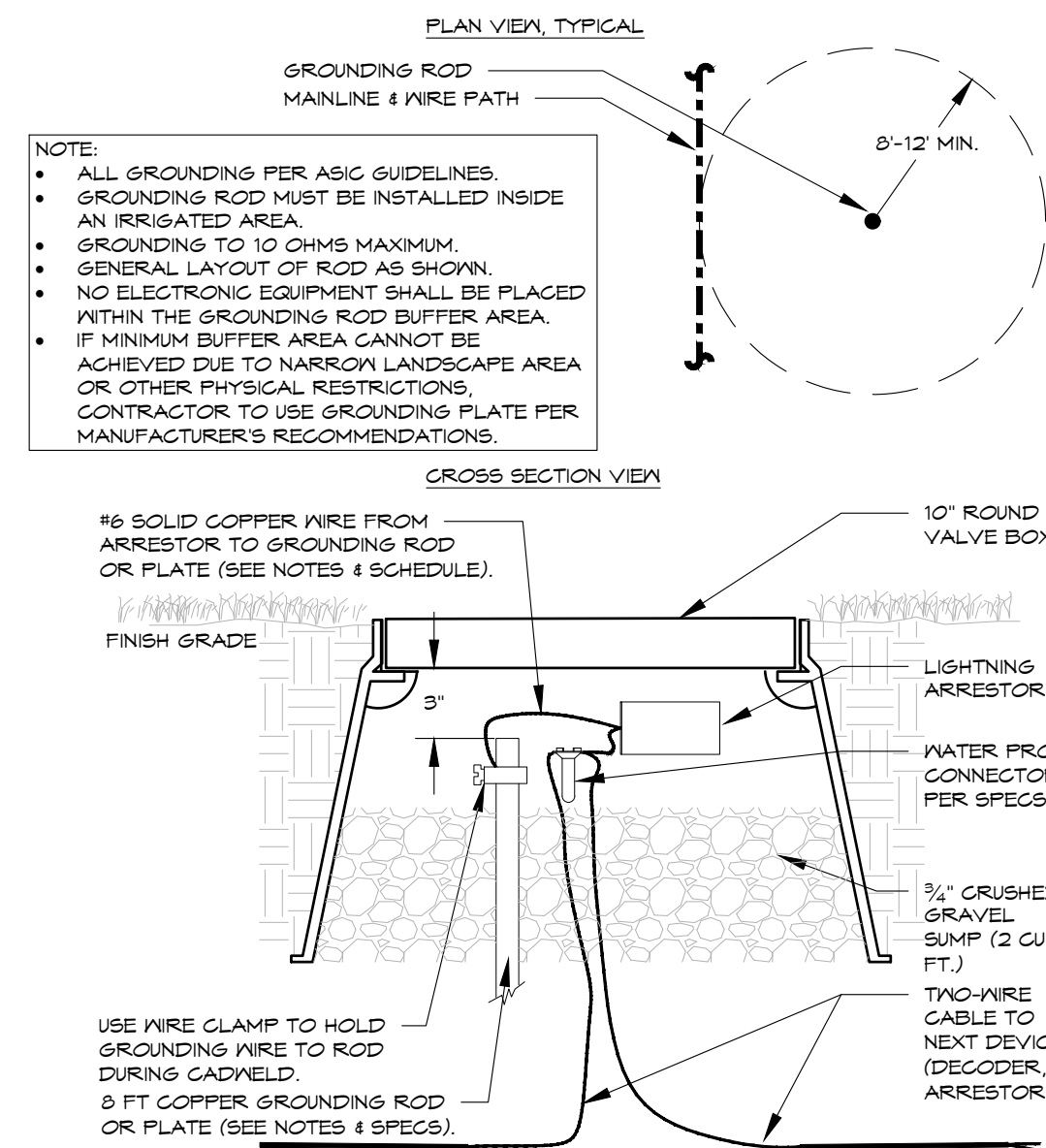
BUBBLER
WITH CHECK VALVE - PVC

13



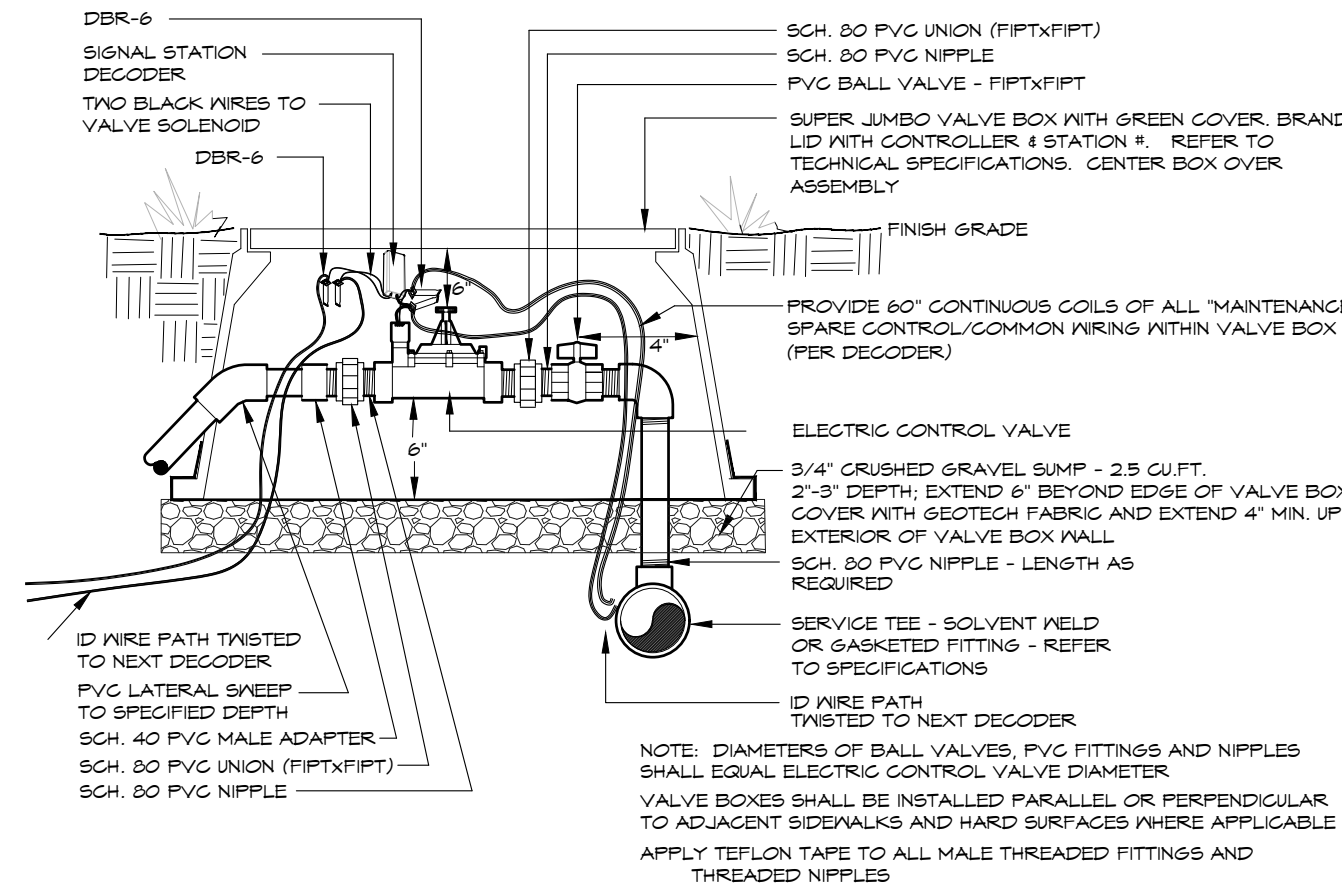
MANUAL DRAIN VALVE
TYPICAL

17



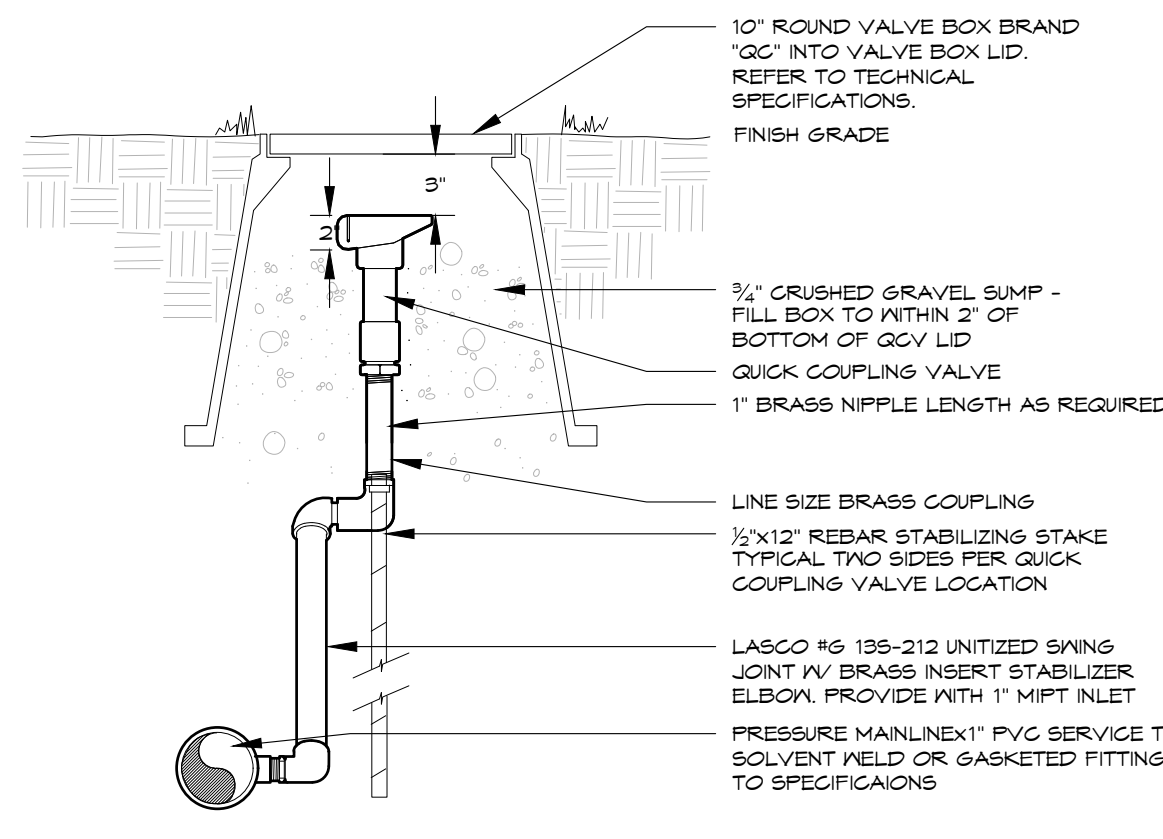
ARRESTOR GROUNDING
TWO-WIRE SYSTEM - TYPICAL

21



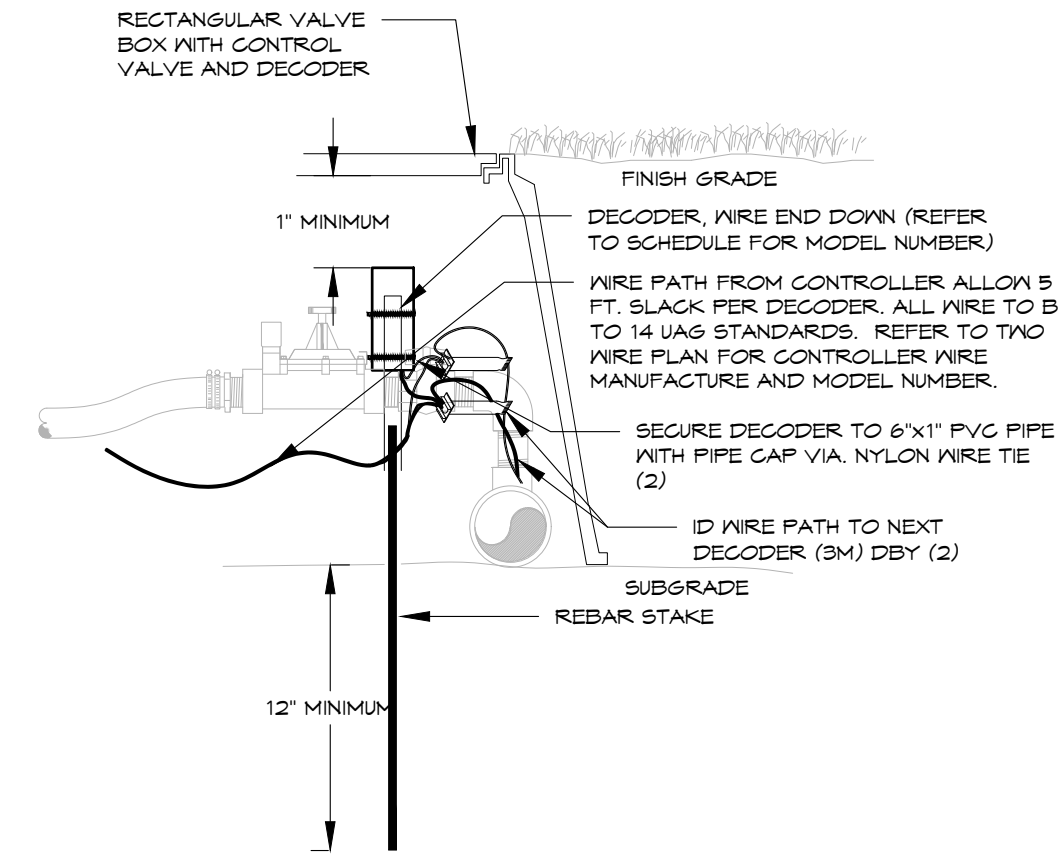
ELECTRIC CONTROL VALVE
WITH DECODER (POTABLE)

14



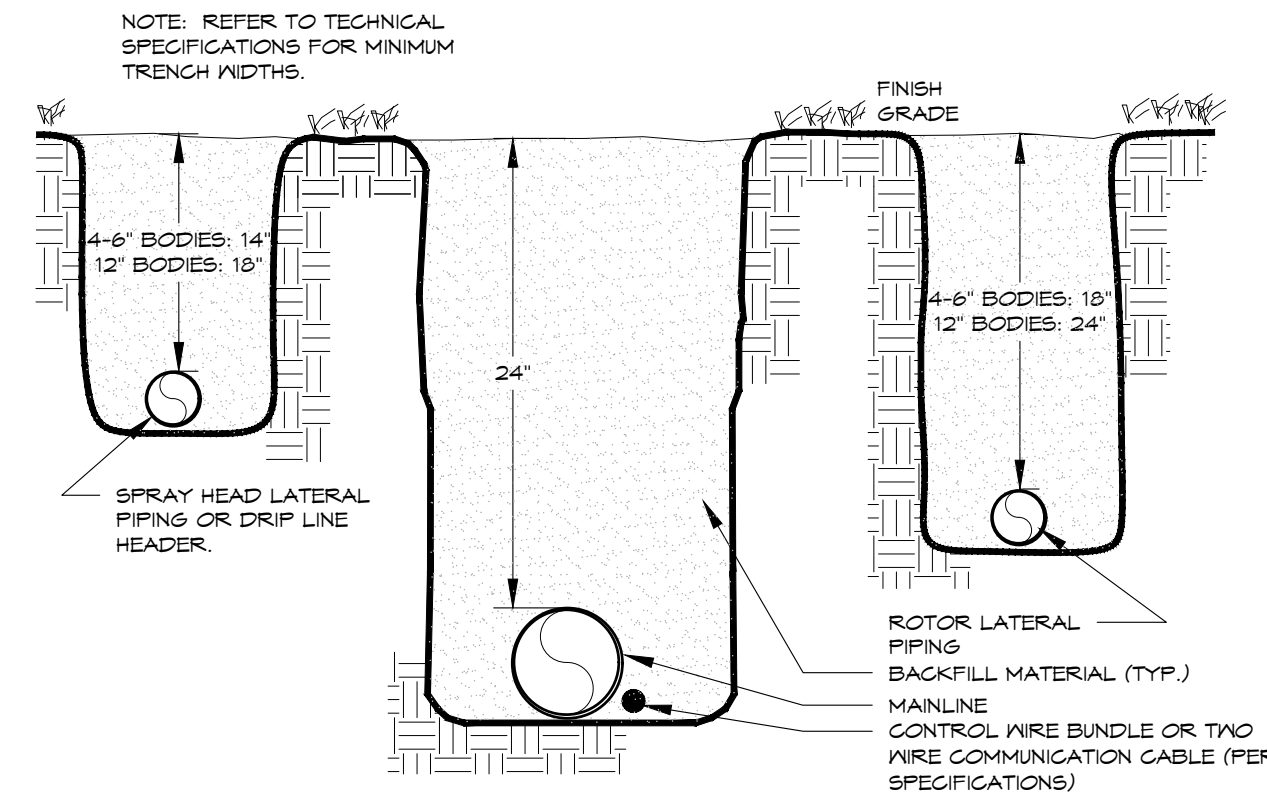
QUICK COUPLING VALVE
LASCOSWING - TYPICAL

18



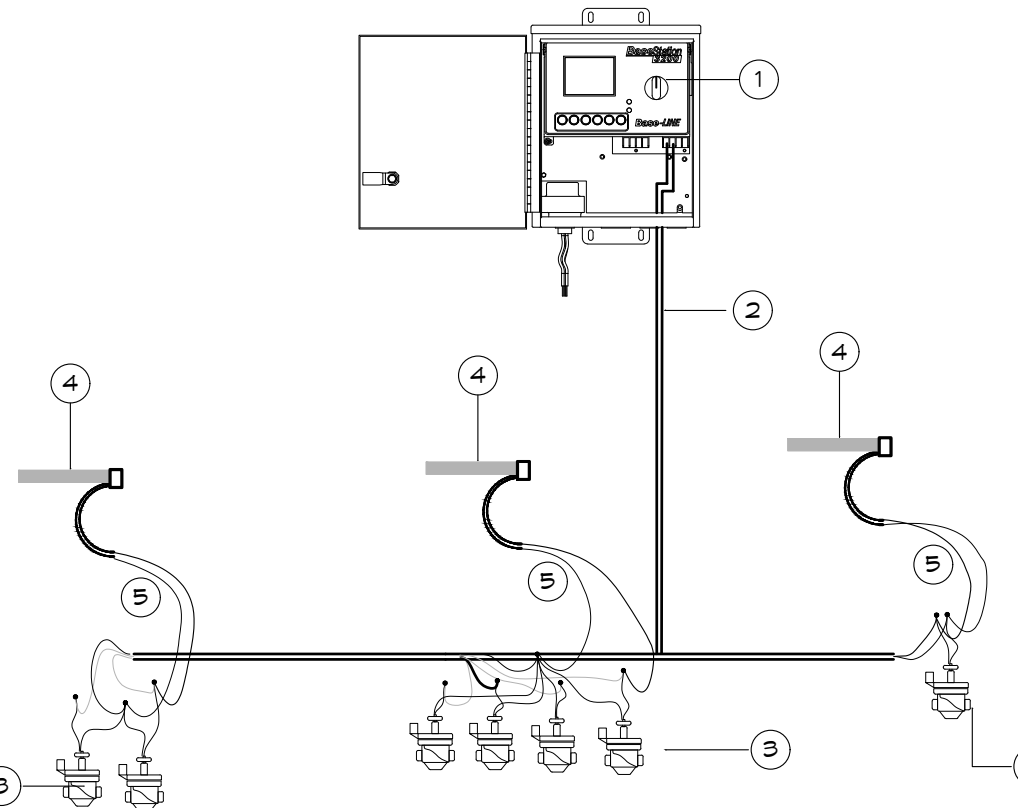
DECODER INSTALLATION
TWO-WIRE SYSTEM

15



TRENCH
24" MAINLINE

19



TYPICAL FIELD CONTROLLER DIAGRAM

22

DIRECTORY
IRRIGATION SCHEDULE
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IRRIGATION DETAILS

IR1.0
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IR1.1 - IR1.2
IR1.3 - IR1.5



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JOB NO. 2309

CITY OF THORNTON
FIRE STATION #8
15343 MONACO STREET
THORNTON, COLORADO 80602

REVISION NAME	DATE

ISSUE DATE: 2024 0927

IRRIGATION DETAILS

IR1.4

- 1

LAWN OR SURFACE TREATMENT

2

6" ROUND JUNCTION BOX, WITH PURPLE LID

3

FINISHED GRADE

4

RECTANGULAR STANDARD OR JUMBO VALVE BOX AS REQUIRED, WITH PURPLE LID

5

BURY DEPTH - BISENSOR

6

SLEEVED WIRE

7

BISENSOR - SET LENGTH HORIZONTALLY AND BLADE IN VERTICAL POSITION WITH WIRES OUT OF BOTTOM

8

SUPPORT BLOCK - 2 REQUIRED

9

LATERAL - SIZE PER PLANS

10

RED TO RED AND BLK TO BLK IN WATER PROOF SPLICE CAP, DBRY OR EQ.

11

24 VOLT REMOTE CONTROL VALVE

12

TWO-WIRE - GAUGE AS PER PLAN

13

12" WIRE BURY OR PER NEG

14

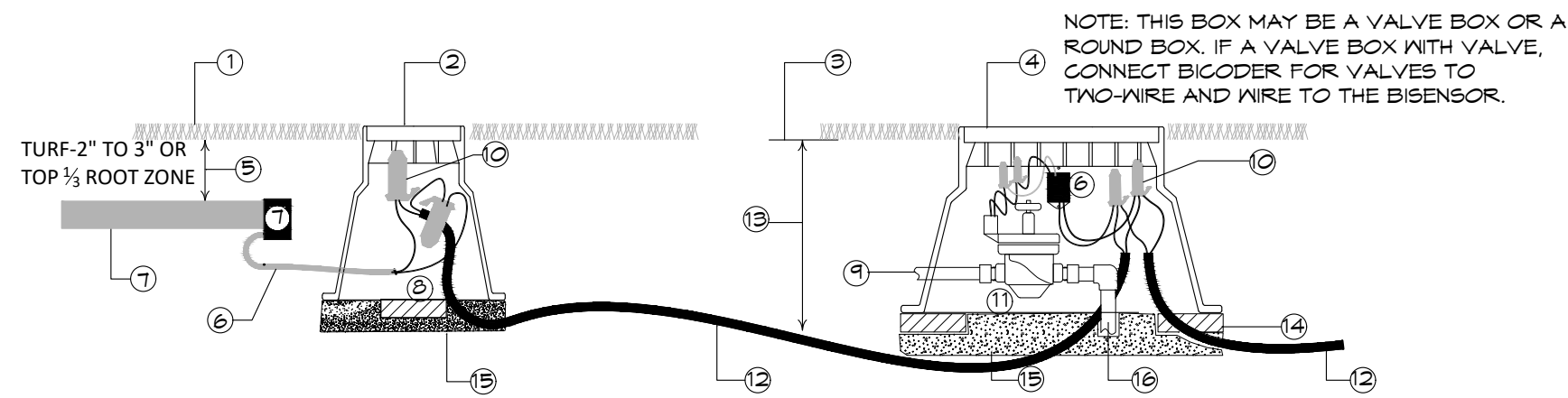
SUPPORT BLOCK - TYPICAL EACH CORNER (4 REQUIRED)

15

GRANULAR MATERIAL FOR DRAINAGE 2" MINIMUM DEPTH

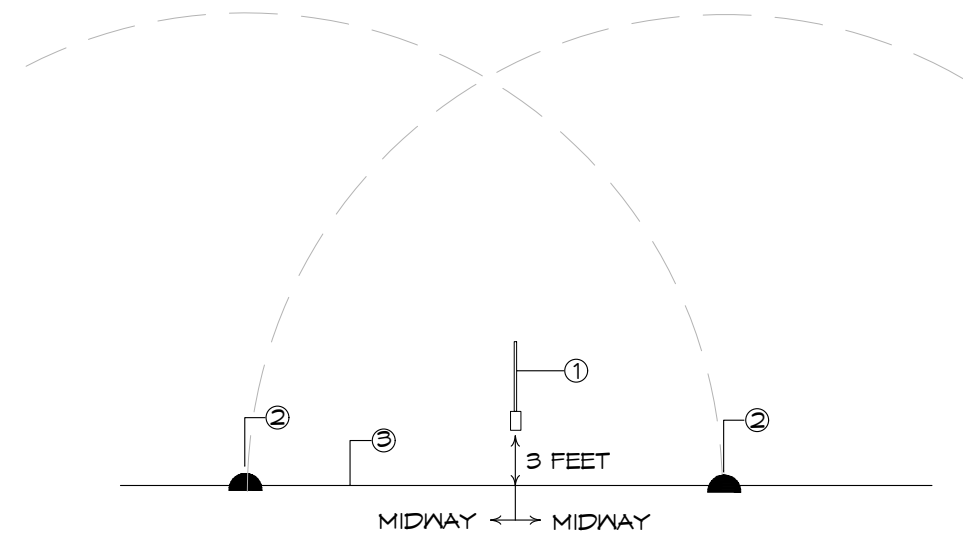
16

PRESSURE LINE - SIZE PER PLAN



SOIL MOISTURE SENSOR

23



- 1

MOISTURE SENSOR
- 2

SPRINKLER HEAD
- 3

EDGE OF TURF

SENSOR LOCATION WITHIN A PART CIRCLE ZONE

24

Project Name: Thornton Fire Station #8					
Project Location: Thornton, Colorado					
Top Description: Top 1					
Meter Sizing Criteria: Public					
Date prepared: 19-Jun-24					
Seasonal Water Use and Peak Flow Projections Prepared By HydroSystems-KDI, Inc.					
	Irrigated Area (Acre)	Month	Application Rate (in/mo/sqft)	Water Application (gallons)	Peak Season Projected Demand (GPM)
Kentucky Bluegrass Spray Irrigated	0.04	April	2.91	2,960	0.8
		May	6.87	6,991	
		June	9.14	9,302	
		July	9.87	10,049	
		August	8.58	8,738	
		September	6.65	6,765	
Irrigation Season Totals			46.45	47,286	
Buffalo Grass Spray Irrigated	0.19			0.15 acre-feet	
		April	1.62	8,187	2.2
		May	3.82	19,338	
		June	5.08	25,731	
		July	5.48	27,798	
		August	4.77	24,172	
September	3.69	18,714			
October	1.35	6,862			
Irrigation Season Totals			25.81	130,802	
Native Tall Grass Spray Irrigated	1.32			0.40 acre-feet	
		April	0.97	34,753	9.3
		May	2.29	82,083	
		June	3.05	109,223	
		July	3.29	117,994	
		August	2.86	102,603	
September	2.22	79,435			
October	0.81	29,126			
Irrigation Season Totals			15.48	555,216	
Shrubs & Trees Spray Irrigated	0.07			1.70 acre-feet	
		April	1.62	3,134	0.8
		May	3.82	7,401	
		June	5.08	9,848	
		July	5.48	10,639	
		August	4.77	9,251	
September	3.69	7,162			
October	1.35	2,626			
Irrigation Season Totals			25.81	50,062	
				0.15 acre-feet	
Site Totals					
Irrigated Area:			70,399 Sq.Ft.	1.62 Acres	
Water Use per Year:			783,366 Gallons	2.40 Acre-feet	
Average Water per Year:			11.13 Gallons/SqFt	17.85 Inches/SqFt	
Peak Season Monthly Water Use:			166,480 Gallons	0.51 Acre-feet	
Average Peak Season Flow:			13 GPM		
Required Meter Size:			1 Inches		
Projections are based on a three day per week, sixteen hour per day, peak season watering schedule.					
Peak season application for Kentucky Bluegrass is 2.23" per week.					
Peak season application for Buffalo Grass is 1.24" per week.					
Peak season application for Native Tall Grass is 0.74" per week.					
Peak season application for Shrubs & Trees is 1.24" per week.					

WATER USE CHART

C1

DIRECTORY

IRRIGATION SCHEDULE	IR1.0
IRRIGATION NOTES	IR1.0
IRRIGATION PLANS	IR1.1 - IR1.2
IRRIGATION DETAILS	IR1.3 - IR1.5

ALLRED & ASSOCIATES

ARCHITECTURE | PLANNING | INTERIOR DESIGN

580 BURBANK STREET, SUITE 125, BROOMFIELD, COLORADO 80020

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• JOB NO. 2309

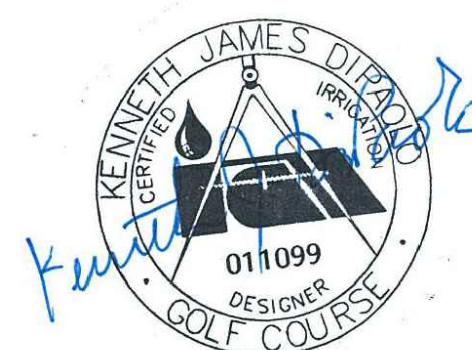
BID SET

CITY OF THORNTON

FIRE STATION #8

15343 MONACO STREET

THORNTON, COLORADO 80602



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IRRIGATION DETAILS

IR1.5



Know what's below.
Call before you dig.
CALL 3 BUSINESS DAYS IN ADVANCE BEFORE YOU
DIG. GRADE, OR EXCAVATE FOR THE MARKING OF
UNDERGROUND MEMBER UTILITIES.



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